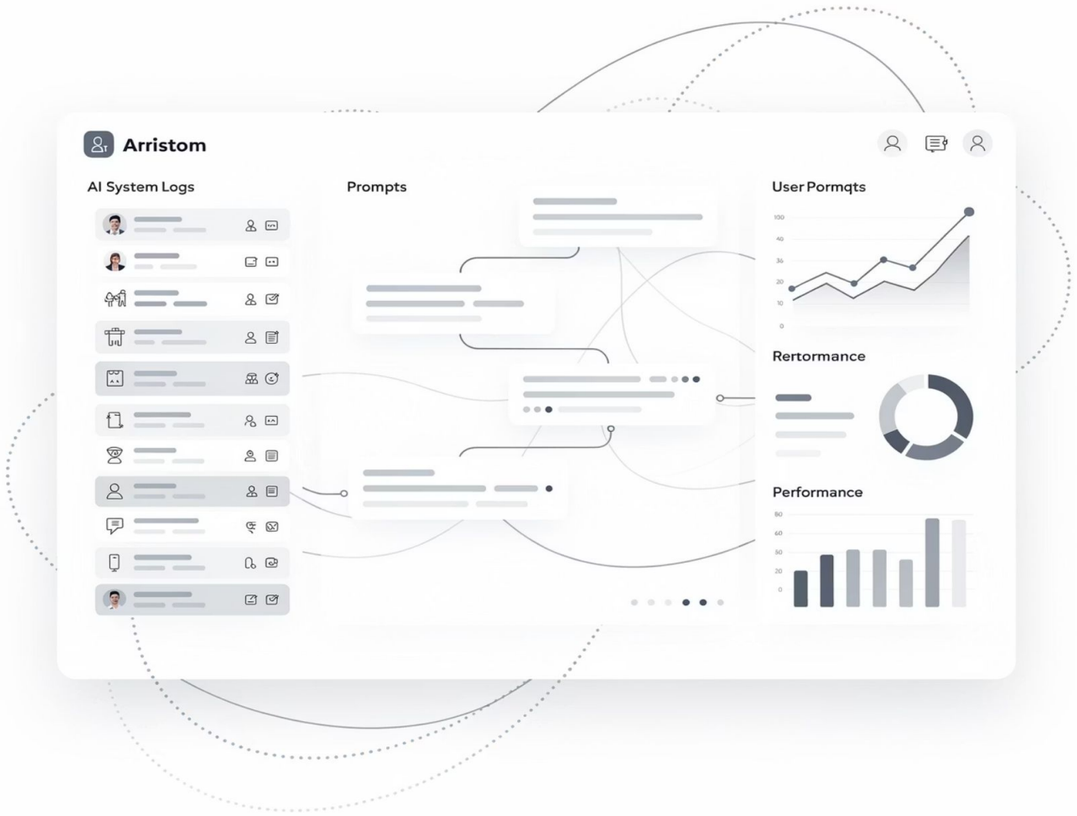
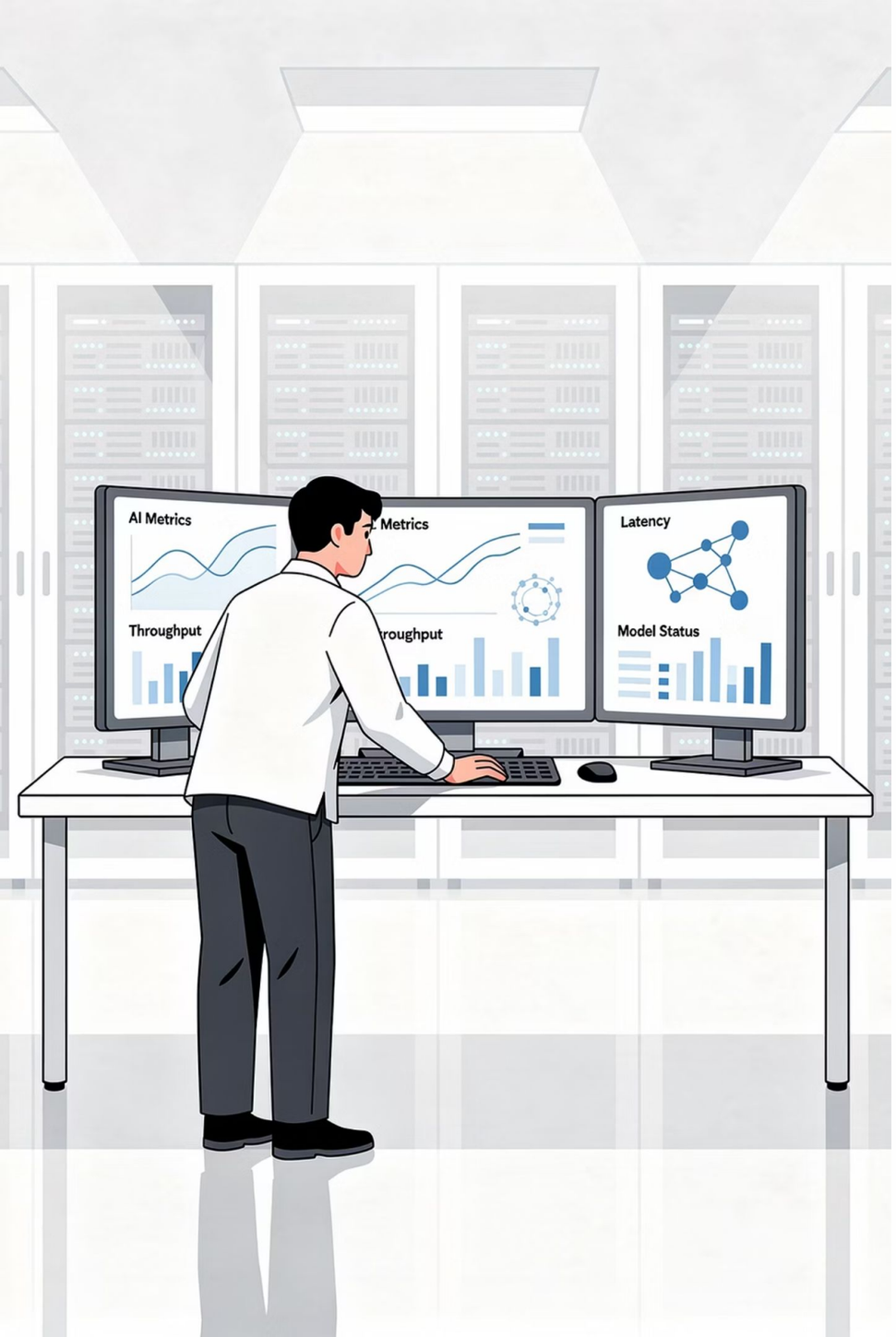
LESSON 3.3.1

Prompt and Log Monitoring

This lesson covers how to monitor and manage prompts and logs in AI systems ensuring performance, compliance, security, and log integrity across your environment.





Purpose of Prompt and Log Monitoring

Prompt and log monitoring requires system administrators to **capture, analyze, and respond** to how AI systems are performing in real time.

Understand System Use

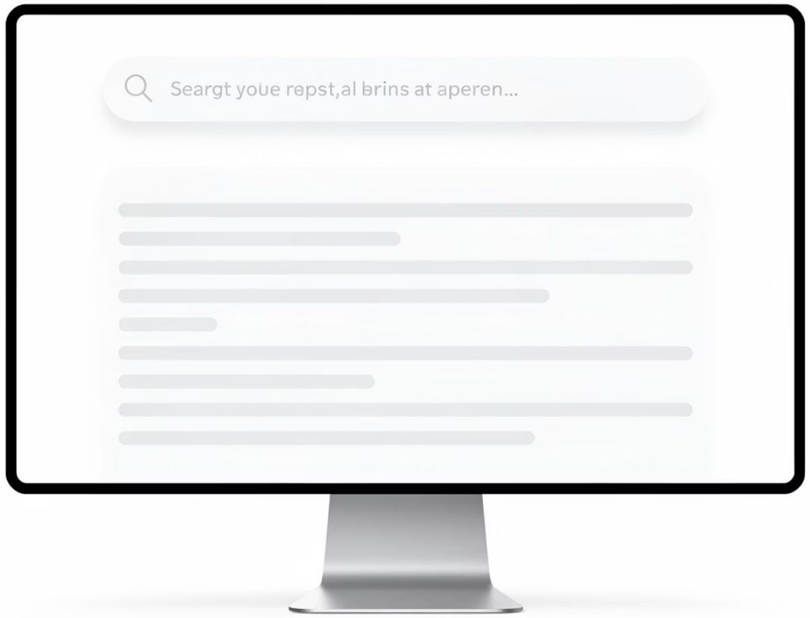
Track usage patterns and identify how the AI is being accessed and by whom.

Ensure Compliance

Maintain alignment with regulatory requirements and acceptable use policies.

Optimize Performance

Identify bottlenecks and tune the system for reliability and efficiency.



PROMPT MONITORING

What to Monitor in Prompts

Effective prompt monitoring tracks both volume and quality of interactions across the AI system.

Query & Response Volume

Count prompts submitted and responses retrieved to establish usage baselines.

Peak Use Times

Identify high-traffic windows to plan hardware scaling for reliable uptime.

Response Accuracy & Compliance

Validate outputs against acceptable use guidelines and policy thresholds.

Inaccurate Response Trends

Rising false or inaccurate responses may signal degraded or outdated training data.

LOG METRICS

Key Log Metrics to Monitor

1

Latency

Measures response delay in milliseconds (ms). High latency signals performance degradation or resource contention.

2

Stop Parameters

Did the user follow up or close the request? Helps evaluate session completion and user satisfaction.

3

Error Messages

Log and classify error codes to identify recurring failures and system instability patterns.

4

Token Use

Track tokens consumed per user. Excessive usage may indicate misuse, abuse, or runaway processes.

5

Unauthorized Access

Flag anomalous data access attempts and potential data exposure events immediately.

SAFETY MONITORING

Monitoring for Unsafe Behavior

AI systems must be actively monitored to detect harmful, policy-violating, or manipulative interactions before they escalate.

Detection Methods: Machine learning pattern matching and keyword searches are the primary tools for automated unsafe behavior detection.

Abusive or Malicious Content

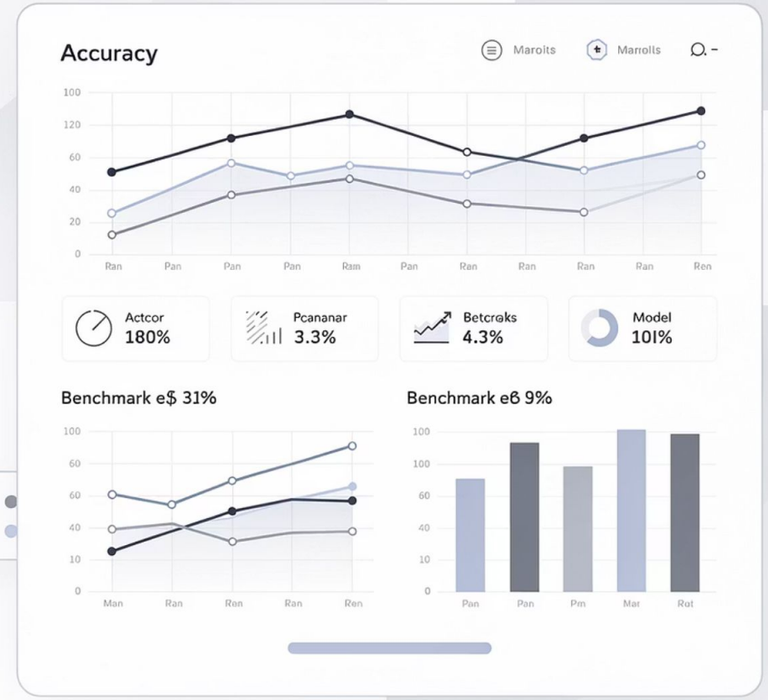
Scan queries and prompts for harmful language, exploitation attempts, or attack patterns.

Sensitive or Prohibited Topics

Flag requests involving restricted subjects as defined by organizational policy.

Contextual Drift

Monitor consecutive queries for gradual topic shifts that attempt to bypass safety guardrails.



MODEL EVALUATION

Model Performance Evaluation

BLEU Score

Bilingual Evaluation Understudy evaluates language model performance, including translation quality, by comparing outputs to a human-curated reference set.

Model Drift

Continuously track and evaluate model outputs over time. When outputs degrade from the baseline, **retrain the model** to re-establish accuracy standards.

Canary Prompts

Pre-formatted test prompts injected to verify that model responses remain **accurate, safe, and within policy limits** — a proactive quality gate.

LOG INTEGRITY

Log Sanitization and Integrity

Protecting the integrity and confidentiality of log data is a critical administrative responsibility.

Sanitize Sensitive Data

Remove PII, PHI, SSNs, credit card numbers, addresses, and phone numbers from all log files.

Read-Only Permissions

Set log files to read-only to prevent unauthorized modification or deletion by users.

Remote Logging

Maintain off-site log copies that are not readily accessible to local users, ensuring a tamper-resistant audit trail.

3.3.1 SUMMARY

Key Takeaways

1

Prompt Monitoring Goals

Identifies peak use, evaluates response accuracy, and ensures policy compliance.

2

Critical Log Metrics

Track latency, stop parameters, error codes, and token use per user.

3

Unsafe Behavior Detection

Monitor for malicious content, prohibited topics, and contextual drift via ML pattern matching.

4

BLEU & Canary Prompts

BLEU evaluates translation quality; canary prompts actively test response accuracy and safety.

5

Model Drift Management

Track output degradation over time and retrain models to restore baseline performance.

6

Log Protection

Sanitize PII/PHI, enforce read-only permissions, and use remote logging to maintain integrity.